
Forecasting & Time Series Analysis

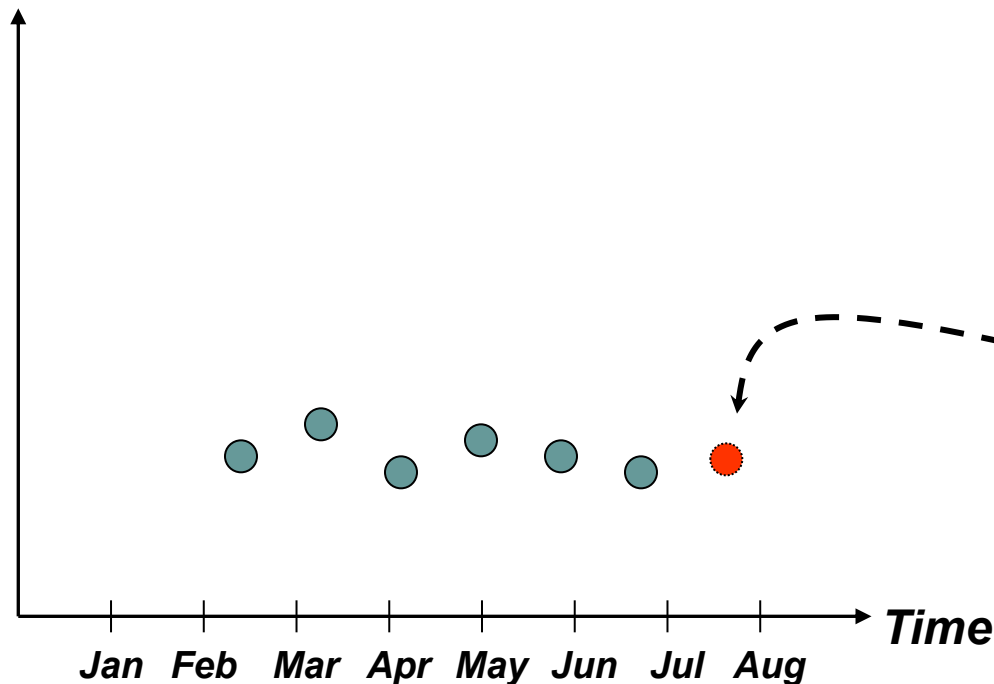
What is Forecasting?

- ☑ ***Forecasting is a tool used for predicting future demand based on past demand information***
- ☑ ***Educated Guess***
- ☑ ***Underlying basis of all business decisions***



What is forecasting all about?

Demand for Mercedes E Class



We try to predict the future by looking back at the past

Predicted demand looking back six months

- Actual demand (past sales)
- Predicted demand

Facts in Forecasting

- Main assumption: Past pattern repeats itself into the future.
- Forecasts are rarely perfect: Don't expect forecasts to be exactly equal to the actual data.
- The science and art of forecasting try to minimize, but not to eliminate, forecast errors. Forecast errors mean the difference between actual and forecasted values.
- Forecasts for a group of products are usually more accurate than these for individual products; a shorter period tend to be more accurate.
- Computer and IT are critical parts of the modern forecasting in large corporations.

Introduction to Time Series Analysis

- A *time-series* is a set of observations on a quantitative variable collected over time.
- Examples
 - continuous monitoring of a person's heart rate
 - hourly readings of air temperature
 - daily closing price of a company stock
 - monthly rainfall data
 - yearly sales figures
- Businesses are often very interested in forecasting time series variables.
- In time series analysis, we analyze the past behavior of a variable in order to predict its future behavior.

We explain it from the following example:

Day	No. of Packets of milk sold
Monday	90
Tuesday	88
Wednesday	85
Thursday	75
Friday	72
Saturday	90
Sunday	102

Year	Population (in Million)
1921	251
1931	279
1941	319
1951	361
1961	439
1971	548
1981	685

- From example 1 it is clear that the sale of milk packets is decrease from Monday to Friday then again its start to increase.
- Same thing in example 2 the population is continuously increase.

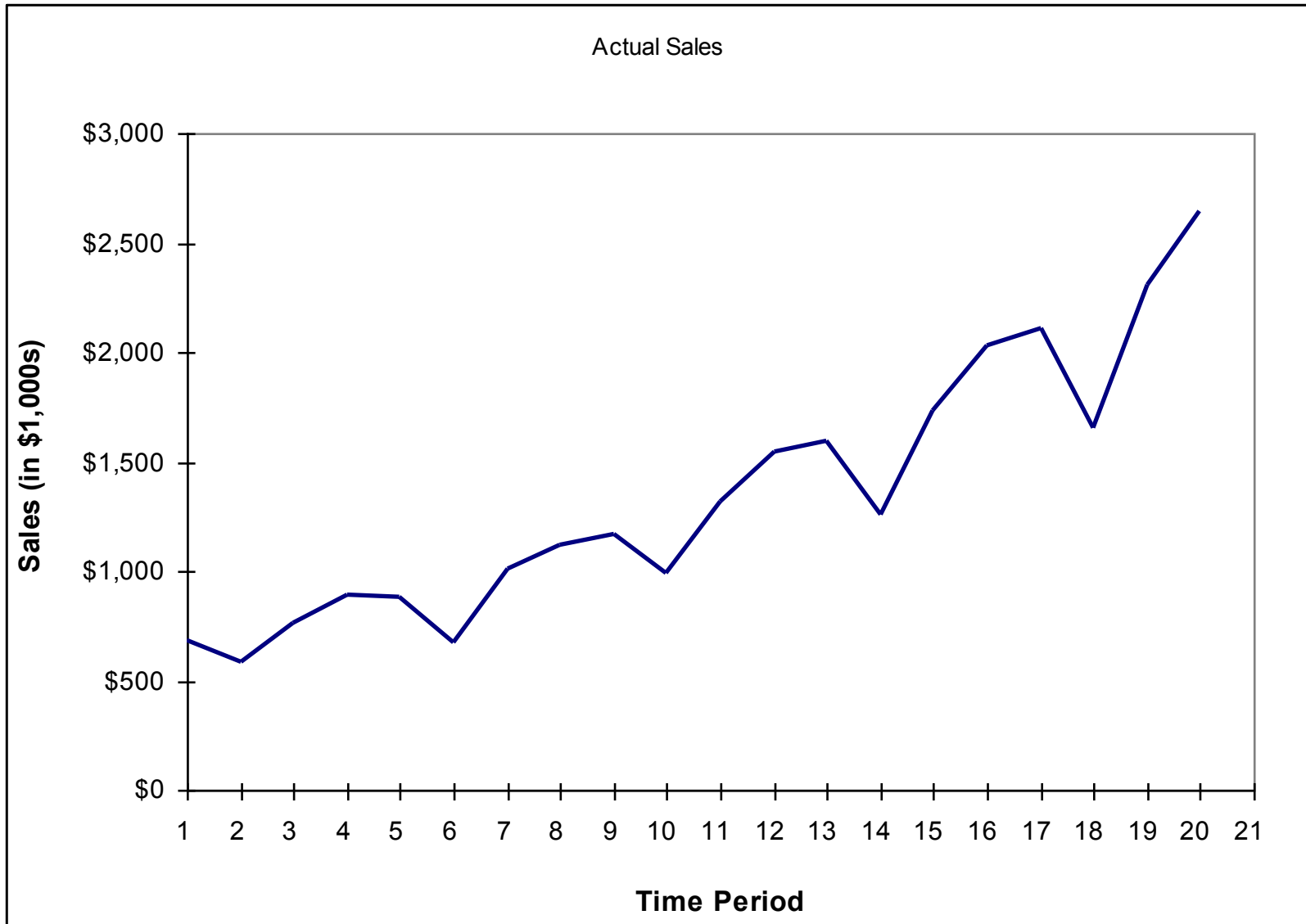
Methods used in Forecasting

- Regression Analysis
- Time Series Analysis (TSA)
 - A statistical technique that uses time-series data for explaining the past or forecasting future events.
 - The prediction is a function of time (days, months, years, etc.)
 - No *causal* variable; examine past behavior of a variable and attempt to predict future behavior

Components of TSA

- Time Frame (How far can we predict?)
 - short-term (1 - 2 periods)
 - medium-term (5 - 10 periods)
 - long-term (12+ periods)
 - No line of demarcation
- Trend
 - Gradual, long-term movement (up or down) of demand.
 - Easiest to detect

Time Series Plot



Some Time Series Terms

- Stationary Data - a time series variable exhibiting no significant upward or downward trend over time.
- Nonstationary Data - a time series variable exhibiting a significant upward or downward trend over time.
- Seasonal Data - a time series variable exhibiting a repeating patterns at regular intervals over time.

Approaching Time Series Analysis

- There are many, many different time series techniques.
- It is usually impossible to know which technique will be best for a particular data set.
- It is customary to try out several different techniques and select the one that seems to work best.
- To be an effective time series modeler, you need to keep several time series techniques in your “tool box.”

Components of Time Series:-

The change which are being in time series, they are effected by Economic, Social, Natural, Industrial & Political Reasons. These reasons are called components of Time Series.

- ❑ SECULAR TREND :-
- ❑ SEASONAL VARIATION :-
- ❑ CYCLICAL VARIATION :-
- ❑ IRREGULAR VARIATION :-

□ Secular trend:

The increase or decrease in the movements of a time series is called Secular trend.

A time series data may show upward trend or downward trend for a period of years and this may be due to factors like:

- increase in population,
- change in technological progress ,
- large scale shift in consumers demands,

For example,

- population increases over a period of time, price increases over a period of years, production of goods on the capital market of the country increases over a period of years. These are the examples of upward trend.
- The sales of a commodity may decrease over a period of time because of better products coming to the market. This is an example of declining trend or downward.

❑ Seasonal variation:

- Seasonal variation are short-term fluctuation in a time series which occur periodically in a year. This continues to repeat year after year.
 - The major factors that are weather conditions and customs of people.
 - More woollen clothes are sold in winter than in the season of summer .
 - Each year more ice creams are sold in summer and very little in Winter season.
 - The sales in the departmental stores are more during festive seasons that in the normal days.

Cyclical Variations:

Cyclical variations are recurrent upward or downward movements in a time series but the period of cycle is greater than a year. Also these variations are not regular as seasonal variation.

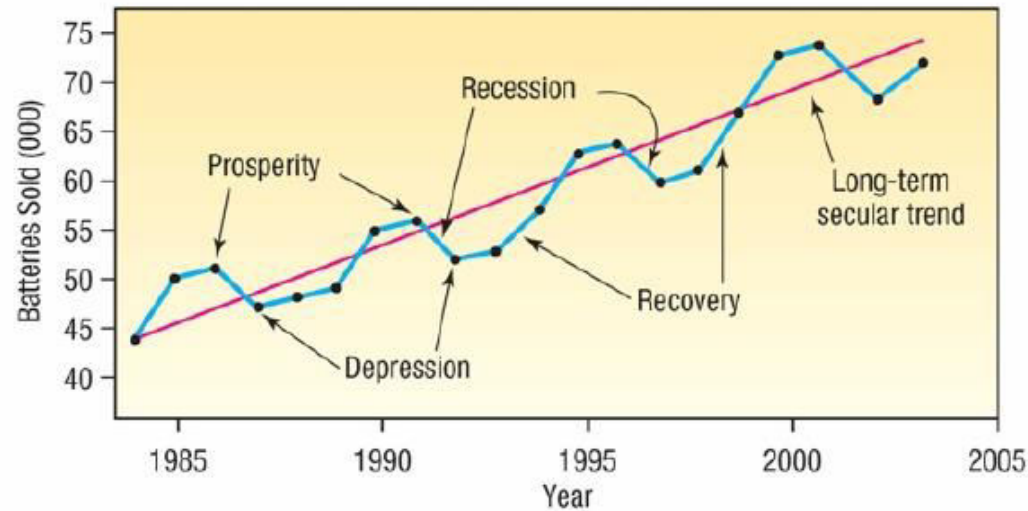


CHART 16-1 Batteries Sold by National Battery Retailers, Inc., from 1984 to 2004

A business cycle showing these oscillatory movements has to pass through four phases - prosperity, recession, depression and recovery. In a business, these four phases are completed by passing one to another in this order.

❑ Irregular variation:

Irregular variations are fluctuations in time series that are short in duration, erratic in nature and follow no regularity in the occurrence pattern. These variations are also referred to as residual variations since by definition they represent what is left out in a time series after trend, cyclical and seasonal variations. Irregular fluctuations results due to the occurrence of unforeseen events like :

- **FLOODS,**
- **EARTHQUAKES,**
- **WARS,**
- **FAMINES**

□ Time Series Model

- Addition Model:

$$Y = T + S + C + I$$

Where:- Y = Original Data

T = Trend Value

S = Seasonal Fluctuation

C = Cyclical Fluctuation

I = Irregular Fluctuation

- Multiplication Model:

$$Y = T \times S \times C \times I$$

or

$$Y = TSCI$$

❑ Measurement of Secular trend:-

- The following methods are used for calculation of trend:
 - ❑ FREE HAND CURVE METHOD:
 - ❑ MOVING AVERAGE METHOD:
 - ❑ LEAST SQUARE METHOD:
 - ❑ EXPONENTIAL SMOOTHING METHOD:

Free hand Curve Method:-

- In this method the data is denoted on graph paper. We take “Time” on ‘x’ axis and “Data” on the ‘y’ axis. On graph there will be a point for every point of time. We make a smooth hand curve with the help of this plotted points.

□ Example:

Draw a free hand curve on the basis of the following data:

Years	1989	1990	1991	1992	1993	1994	1995	1996
Profit (in '000)	148	149	149.5	149	150.5	152.2	153.7	153



Moving Average Method

- ☑ Assumes an average is a good estimator of future behavior
- ☑ MA is a series of arithmetic means
- ☑ Used if little or no trend
- ☑ Used often for smoothing
 - ☑ Provides overall impression of data over time

$$\text{Moving average} = \frac{\sum \text{demand in previous } n \text{ periods}}{n}$$

□ Moving Average Method:-

- It is one of the most popular method for calculating Long Term Trend. This method is also used for 'Seasonal fluctuation', 'cyclical fluctuation' & 'irregular fluctuation'. In this method we calculate the 'Moving Average for certain years.
- For example: If we calculating 'Three year's Moving Average' then according to this method:

$$= \frac{(1)+(2)+(3)}{3}, \quad \frac{(2)+(3)+(4)}{3}, \quad \frac{(3)+(4)+(5)}{3}, \quad \dots\dots\dots$$

Where (1),(2),(3),..... are the various years of time series.

□ Example: Find out the 3 year's moving Average:

Year	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Price	20	25	33	33	27	35	40	43	35	32	37	48	50	37	45

Year (1)	Price of sugar (Rs.) (2)	Three year's moving Total (3)	Three year's moving Average (Col 3/3) (4)
1982	20	-	-
1983	25	-	-
1984	33	$(20+25+33)=78$	26
1985	30	$(25+33+30)=88$	29
1986	27	90	30
1987	35	And so on	
1988	40		
1989	43		
1990	35		
1991	32		
1992	37		
1993	48		
1994	50		
1995	37	-	-
1996	45	-	-

❑ Least Square Method:-

- This method uses the given series of data to develop a trend line for predictive purposes
- The least-squares method establishes a trend line from;

- $Y_t = a + bx$ where $a = \frac{\sum y}{n}$

$$b = \frac{\sum xy}{\sum x^2}$$

□ Least Square Method Example:-

- Annual soft drink sales
- Find the expected sales for 2001

Year	Y	[x]	x ²	xy
1991	13	-4	16	-52
1992	15	-3	9	-45
1993	17	-2	4	-34
1994	18	-1	1	-18
1995	19	0	0	0
1996	20	1	1	20
1997	20	2	4	40
1998	21	3	9	63
1999	22	4	16	88
	165		60	62

$$a = \frac{\sum y}{n} \quad b = \frac{\sum xy}{\sum x^2}$$

$$a = \frac{165}{9} \quad b = \frac{62}{60}$$

$$a = 18.3 \quad b = 1.03$$

$$Y_t = 18.3 + 1.03x$$

$$2001 \ Y_t = 18.3 + 1.03(6)$$

$$= 18.3 + 6.18$$

$$= 22.48$$

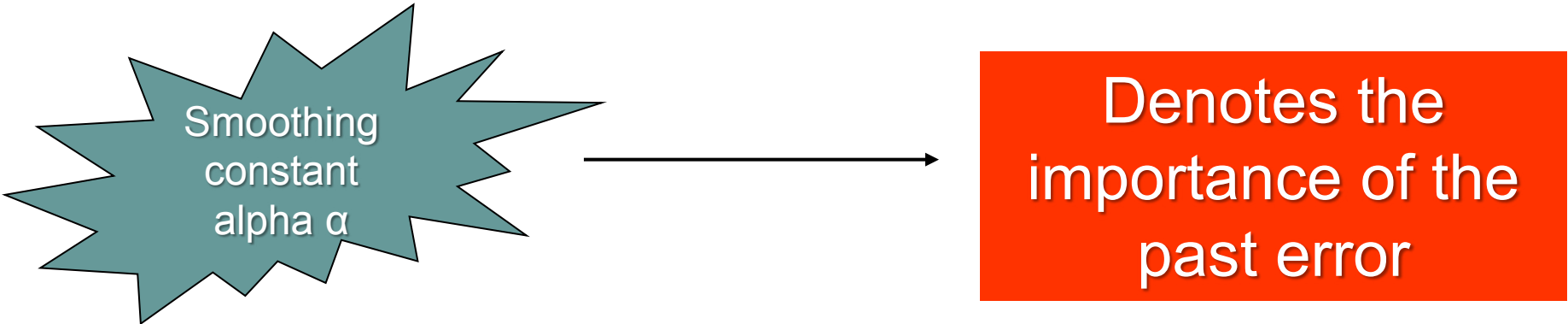
Expected sales for 2001 = \$22,480,000

Y is the given data

X is the year value in relation to the middle year

Exponential Smoothing

Main idea: The prediction of the future depends mostly on the most recent observation, and on the error for the latest forecast



Smoothing
constant
alpha α

Denotes the
importance of the
past error

Exponential Smoothing

***New forecast = Last period's forecast
+ α (Last period's actual demand
– Last period's forecast)***

$$F_{t+1} = F_t + \alpha(S_t - F_t)$$

where F_{t+1} = new forecast

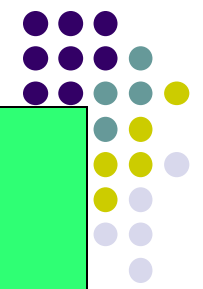
F_t = previous forecast

S_t = previous actual

***α = smoothing (or weighting)
constant ($0 \leq \alpha \leq 1$)***

Exponential Smoothing

- ☑ Assumes the most recent observations have the highest predictive value
- ☑ Requires smoothing constant (α)
 - ☑ Ranges from 0 to 1
 - ☑ Subjectively chosen
- ☑ Involves little record keeping of past data



Exponential Smoothing Example

Predicted demand = 142 Ford Mustangs

Actual demand = 153

Smoothing constant $\alpha = .20$

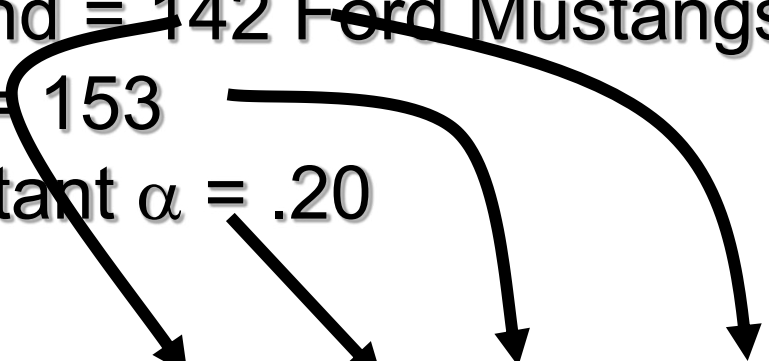
Exponential Smoothing Example

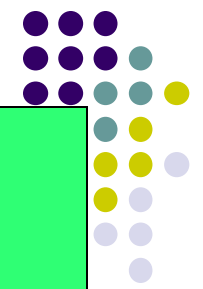
Predicted demand = 142 Ford Mustangs

Actual demand = 153

Smoothing constant $\alpha = .20$

New forecast = $142 + .2(153 - 142)$





Exponential Smoothing Example

Predicted demand = 142 Ford Mustangs

Actual demand = 153

Smoothing constant $\alpha = .20$

$$\begin{aligned}\text{New forecast} &= 142 + .2(153 - 142) \\ &= 142 + 2.2 \\ &= 144.2 \approx 144 \text{ cars}\end{aligned}$$

Forecasting Software

☑ Spreadsheets

- ☑ Microsoft Excel, Quattro Pro, Lotus 1-2-3
- ☑ Limited statistical analysis of forecast data

☑ Statistical packages

- ☑ SPSS, SAS, NCSS, Minitab
- ☑ Forecasting plus statistical and graphics

☑ Specialty forecasting packages

- ☑ Forecast Master, Forecast Pro, Autobox, SCA

How to do it in excel